

Lecture Note 10: Large Sample Tests

14.380: Statistical Methods in Economics, Fall 2026

1 Likelihood Ratio Test

Let $X_1, \dots, X_n$ be a random sample from a distribution with pdf $f(x|\theta)$ where θ is some one dimensional (unknown) parameter. Suppose we want to test the null hypothesis, H_0 , that $\theta = \theta_0$ against the alternative hypothesis, H_a , that $\theta \neq \theta_0$. Assume the same regularity conditions hold as in the MLE theory. Then the likelihood ratio test (LRT) statistic is

$$\lambda(x) = \frac{\mathcal{L}(\theta_0|x)}{\mathcal{L}(\hat{\theta}_{ML}|x)}$$

where $x = (x_1, \dots, x_n)$ is a realization of the data set and $\hat{\theta}_{ML}$ is the ML estimator. Then we have

Theorem 1. *Under the same regularity conditions as in the MLE theory and if $H_0 : \theta = \theta_0$ holds, we have:*

$$-2 \log \lambda(X) \xrightarrow{d} \chi_1^2.$$

Proof. Denote $\ell(\theta|x) = \log \mathcal{L}(\theta|x)$. By the Taylor theorem, for some θ^* between θ_0 and $\hat{\theta}_{ML}$,

$$\begin{aligned} -2 \log \lambda(X) &= -2(\ell(\theta_0|X) - \ell(\hat{\theta}_{ML}|X)) \\ &= -2 \left(\frac{\partial \ell(\hat{\theta}_{ML}|X)}{\partial \theta} (\theta_0 - \hat{\theta}_{ML}) + \frac{1}{2} \frac{\partial^2 \ell(\theta^*|X)}{\partial \theta^2} (\theta_0 - \hat{\theta}_{ML})^2 \right) \\ &= -\frac{\partial^2 \ell(\theta^*|X)}{\partial \theta^2} (\theta_0 - \hat{\theta}_{ML})^2 \end{aligned}$$

since $\partial \ell(\hat{\theta}_{ML}|X)/\partial \theta = 0$ by FOC.

By the MLE theory, $\hat{\theta}_{ML} \rightarrow_p \theta_0$. So, $\theta^* \rightarrow_p \theta_0$. As will be shown in 14.385, by the uniform law of large numbers,

$$-\frac{1}{n} \frac{\partial^2 \ell(\theta^*|X)}{\partial \theta^2} = -\frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \log f(X_i|\theta^*)}{\partial \theta^2} \rightarrow_p I_1(\theta_0)$$

where $I_1(\theta)$ denotes the information for one observation, i.e. $I_1(\theta) = -\mathbb{E}_\theta[\partial^2 \log f(X_i|\theta)/\partial \theta^2]$. By the Slutsky theorem,

$$-\frac{1}{n I_1(\theta_0)} \frac{\partial^2 \ell(\theta^*|X)}{\partial \theta^2} \rightarrow_p 1$$

In addition, from the MLE theory,

$$\sqrt{n}(\hat{\theta}_{ML} - \theta_0) \xrightarrow{d} N(0, I_1^{-1}(\theta_0)).$$

So, by the continuous mapping theorem,

$$I_1(\theta_0)n(\hat{\theta}_{ML} - \theta_0)^2 \xrightarrow{d} \chi_1^2.$$

By the Slutsky theorem,

$$-2 \log \lambda(X) = -\frac{1}{nI_1(\theta_0)} \frac{\partial^2 \ell(\theta^*|X)}{\partial \theta^2} I_1(\theta_0) n(\theta_0 - \hat{\theta}_{ML})^2 \xrightarrow{d} \chi_1^2.$$

□

It follows from this theorem that the large sample LR test of level α rejects the null hypothesis if and only if $-2 \log \lambda(x) > \chi_1^2(1 - \alpha)$, where $\chi_1^2(1 - \alpha)$ denotes the $1 - \alpha$ quantile of χ_1^2 . Note that in finite samples, the size of this test may be greater than α but as the sample size increases, the size will converge to α .

1.1 Formulation for multi-dimensional case

In general, let θ be a multidimensional parameter (say dimensionality is k). Suppose that the null hypothesis Θ_0 can be written in the form $\{\theta \in \Theta : g_1(\theta) = 0, \dots, g_p(\theta) = 0\}$ where $g_1, \dots, g_p$ denote some nonlinear functions of θ . Equations $g_1(\theta) = 0, \dots, g_p(\theta) = 0$ are called restrictions of the model (and $k \geq p$, if $k > p$ then it is a composite hypothesis, if $k = p$ then simple). Assume that restrictions are jointly independent in the sense that we cannot drop any subset of restrictions without changing the set Θ_0 . Then, under some regularity conditions (mainly smoothness of $g_1, \dots, g_p$),

$$-2 \log \lambda(X) = 2 \left(\max_{\theta \in \Theta} \ell(\theta|X) - \max_{\theta \in \Theta_0} \ell(\theta|X) \right) \xrightarrow{d} \chi_p^2$$

under the assumption that the null hypothesis holds. So, the large sample LR test of level α rejects the null hypothesis if and only if $-2 \log \lambda(X) > \chi_p^2(1 - \alpha)$. Often, we denote $LR = -2 \log \lambda(X)$. LR is called the likelihood ratio statistic. Let us denote $\hat{\theta}_0 = \arg \max_{\theta \in \Theta_0} \ell(\theta|X)$ to be the restricted estimate (the estimate assuming the null is true), then

$$LR = 2(\ell(\hat{\theta}_{ML}|X) - \ell(\hat{\theta}_0|X))$$

Example Let $X_1, \dots, X_n$ be a random sample from a $\text{Poisson}(\lambda)$ distribution. Recall that the pmf of the $\text{Poisson}(\lambda)$ distribution is $f(x|\lambda) = \lambda^x e^{-\lambda} / x!$ for $x = 0, 1, 2, \dots$. Suppose we want to test the null hypothesis, H_0 , that $\lambda = \lambda_0 = 6$ against the alternative hypothesis, H_a , that $\lambda \neq \lambda_0$. Suppose we observe $\bar{X}_n = 5$ while our sample size is $n = 100$. Let us derive the result of the large sample LR test. The likelihood function is

$$\mathcal{L}(\lambda|X) = \frac{\lambda^{\sum_{i=1}^n X_i} e^{-n\lambda}}{\prod_{i=1}^n X_i!}$$

where $X = (X_1, \dots, X_n)$. The log-likelihood is

$$\ell(\lambda|X) = \sum_{i=1}^n X_i \log \lambda - n\lambda - \log \prod_{i=1}^n X_i!$$

So, the ML estimator $\hat{\lambda}_{ML}$ solves

$$\sum_{i=1}^n X_i / \hat{\lambda}_{ML} - n = 0$$

or, equivalently,

$$\hat{\lambda}_{ML} = \bar{X}_n$$

So, the LRT statistic is

$$\lambda(X) = (\lambda_0 / \hat{\lambda}_{ML})^{\sum_{i=1}^n X_i} e^{-n(\lambda_0 - \hat{\lambda}_{ML})}.$$

Then

$$\begin{aligned} LR &= -2 \log \lambda(X) \\ &= -2 \left(\sum_{i=1}^n X_i \log(\lambda_0 / \hat{\lambda}_{ML}) - n(\lambda_0 - \hat{\lambda}_{ML}) \right) \\ &= -2n(\bar{X}_n \log(\lambda_0 / \bar{X}_n) - \lambda_0 + \bar{X}_n) \\ &= -200(5 \log(6/5) - 6 + 5) \\ &\approx 17.7, \end{aligned}$$

while $\chi_1^2(0.95) = 3.84$. So the large sample LR test rejects the null hypothesis.

2 Large Sample Tests: Wald

2.1 Simplistic 1-dimensional case

Once we know the asymptotic distribution of some statistic, say, $\delta(X_1, \dots, X_n)$, we can construct a large sample test based on this asymptotic distribution. Suppose we can show that

$$\sqrt{n}(\delta(X_1, \dots, X_n) - \tau) \xrightarrow{d} N(0, \sigma^2)$$

where τ is some 1-dimensional parameter. Suppose we have a consistent estimator $\hat{\sigma}^2$ of σ^2 , i.e. $\hat{\sigma}^2 \rightarrow_p \sigma^2$. By the Slutsky theorem,

$$\sqrt{n}(\delta(X_1, \dots, X_n) - \tau) / \hat{\sigma} \xrightarrow{d} N(0, 1)$$

Suppose we want to test the null hypothesis, H_0 , that $\tau = \tau_0$ against the two-sided alternative. Under the null hypothesis,

$$\sqrt{n}(\delta(X_1, \dots, X_n) - \tau_0) / \hat{\sigma} \xrightarrow{d} N(0, 1).$$

So, one test of level α will be to reject the null hypothesis if $t = \sqrt{n} \frac{\delta(X_1, \dots, X_n) - \tau_0}{\hat{\sigma}}$ is smaller than $z_{\alpha/2}$ or larger than $z_{1-\alpha/2}$. This is equivalent to calculating the statistic

$$W = n \left(\frac{\delta(X) - \tau_0}{\hat{\sigma}} \right)^2$$

and comparing it to the $1 - \alpha$ quantile of the χ_1^2 distribution.

2.2 Multi-dimensional case

Notice that this logic could be easily extended to multi-dimensional parameters. Assume that τ is p -dimensional and

$$\sqrt{n}(\delta(X) - \tau) \xrightarrow{d} N(0, \Sigma),$$

and we can construct a consistent estimate $\hat{\Sigma}$ of the covariance matrix Σ , that is $\hat{\Sigma} \rightarrow_p \Sigma$, then

$$W = n(\delta(X) - \tau_0)' \hat{\Sigma}^{-1} (\delta(X) - \tau_0) \xrightarrow{d} \chi_p^2$$

if $H_0 : \tau = \tau_0$.

Example 1. The multi-dimensional Wald test has a canonical empirical application: checking *covariate balance* in a randomized experiment. Suppose n units are randomly assigned to treatment ($n_T = \gamma n$ units) or control ($n_C = (1 - \gamma)n$ units), and for each unit we observe a vector of characteristics $X_i \in \mathbb{R}^p$ measured *before* treatment — age, income, baseline outcomes. Because assignment is random, treatment carries no information about the characteristics: the population means of X_i are identical in the two groups, so the estimand

$$\delta = \mathbb{E}[X_i \mid D_i = 1] - \mathbb{E}[X_i \mid D_i = 0]$$

equals zero. This is a testable implication of successful randomization. The natural estimator is the vector of differences in sample means, $\hat{\delta} = \bar{X}_T - \bar{X}_C$, and by the central limit theorem applied within each group,

$$\sqrt{n}(\hat{\delta} - \delta) \xrightarrow{d} N(0, V), \quad V = \frac{\Sigma}{\gamma} + \frac{\Sigma}{1 - \gamma},$$

where $\Sigma = \text{Var}(X_i)$ is common to the two groups under randomization. The Wald statistic with a consistent estimator $\hat{V}$ is

$$W = n \hat{\delta}' \hat{V}^{-1} \hat{\delta} \xrightarrow{d} \chi_p^2 \quad \text{under } H_0 : \delta = 0,$$

and we reject at level α if W exceeds the $(1 - \alpha)$ -quantile of χ_p^2 .

Example 2. A second application of the multi-dimensional Wald test arises in *difference-in-differences* designs. Suppose a treated group and a comparison group are each observed over several periods *before* the treatment of interest occurs, with group-level mean outcomes $\bar{y}_{T,t}$ and $\bar{y}_{C,t}$ in pre-treatment periods $t = 0, 1, \dots, p$. The design's identifying idea is that, absent treatment, the

two groups' outcomes move in parallel. That idea has a testable implication in the pre-treatment data: the “placebo” difference-in-differences estimates

$$\hat{\delta}_t = (\bar{y}_{T,t} - \bar{y}_{T,0}) - (\bar{y}_{C,t} - \bar{y}_{C,0}), \quad t = 1, \dots, p,$$

estimate effects of a treatment that has not yet happened, and so should be indistinguishable from zero. Stacking them into $\hat{\delta} \in \mathbb{R}^p$, a central limit theorem for the underlying group-period means gives $\sqrt{n}(\hat{\delta} - \delta) \xrightarrow{d} N(0, V)$ — note that V is not diagonal, since every $\hat{\delta}_t$ shares the same base-period means — and the joint hypothesis of no pre-trends, $H_0: \delta = 0$, is tested with

$$W = n \hat{\delta}' \hat{V}^{-1} \hat{\delta} \xrightarrow{d} \chi_p^2.$$

This “pre-trend test” is reported routinely in empirical work using difference-in-differences. A rejection casts doubt on the parallel-movement idea. A non-rejection, however, should be read with the lesson of Lecture Note 8's power analysis in mind: failing to reject is reassuring only insofar as the test had power to detect pre-trends of a relevant magnitude; Roth (2022) develops this point and its consequences for inference after pre-testing.

2.3 Special case: 1-dimensional MLE

We can specialize this to the MLE case, for example, and see how this test compares to the LR introduced before. Let $\hat{\theta}_{ML}$ be the ML estimator of a 1-dimensional parameter $\theta \in \mathbb{R}$. We know that, under some regularity conditions,

$$\sqrt{n}(\hat{\theta}_{ML} - \theta) \xrightarrow{d} N(0, I_1^{-1}(\theta))$$

Under some regularity conditions, $I_1^{-1}(\theta)$ may be consistently estimated by $I_1^{-1}(\hat{\theta}_{ML})$. Suppose that our null hypothesis is $H_0: \theta = \theta_0$. Then, under the null hypothesis,

$$\sqrt{n}I_1^{1/2}(\hat{\theta}_{ML})(\hat{\theta}_{ML} - \theta_0) \xrightarrow{d} N(0, 1)$$

Under the null hypothesis

$$W = nI_1(\hat{\theta}_{ML})(\hat{\theta}_{ML} - \theta_0)^2 \xrightarrow{d} \chi_1^2.$$

Recall that the LR statistic is given by

$$LR = n(\hat{\theta}_{ML} - \theta_0)^2 \left(-\frac{1}{n} \frac{\partial^2 \ell(\theta^*|X)}{\partial \theta^2} \right)$$

where θ^* is between θ_0 and $\hat{\theta}_{ML}$. As in the case of the Wald statistic, under the null hypothesis,

$$LR \xrightarrow{d} \chi_1^2$$

Moreover,

$$W - LR \rightarrow_p 0$$

since $I_1(\hat{\theta}_{ML}) \rightarrow_p I_1(\theta_0)$ and $-(1/n)\partial^2 \ell(\theta^*|X)/\partial \theta^2 \rightarrow_p I_1(\theta_0)$. Thus, LR and Wald statistics are asymptotically equivalent. They are different in finite samples though. In particular, it is known that $W \geq LR$ in the case of normal likelihood.

An advantage of the Wald statistic in comparison with the LR statistic is that it only includes calculations based on the unrestricted estimator $\hat{\theta}_{ML}$. On the other hand, in order to calculate the Wald statistic, we have to estimate the information matrix.

Example (cont.) Let us calculate the Wald statistic in our example with a random sample from the Poisson(λ) distribution. The log-likelihood is

$$\ell(\lambda|X) = \sum_{i=1}^n X_i \log \lambda - n\lambda - \log \prod_{i=1}^n X_i!$$

So,

$$\partial \ell(\lambda|X)/\partial \lambda = \sum_{i=1}^n X_i/\lambda - n$$

and

$$\partial^2 \ell(\lambda|X)/\partial \lambda^2 = -\sum_{i=1}^n X_i/\lambda^2.$$

Thus,

$$I_1(\lambda) = -\mathbb{E} \left[\frac{1}{n} \frac{\partial^2 \ell(\lambda)}{\partial \lambda^2} \right] = \frac{1}{\lambda}.$$

So, the Wald statistic is

$$W = n(\hat{\lambda} - \lambda_0)^2/\hat{\lambda} = 100 \cdot (5 - 6)^2 \cdot (1/5) = 20.$$

So, the test based on the Wald statistic rejects the null hypothesis with an even smaller p-value than the test based on the LR statistic.

3 Score Test

3.1 1-dimensional case

Recall that the score is defined by

$$S(\theta) = \frac{\partial \ell}{\partial \theta}(\theta|X) = \frac{\partial \log \mathcal{L}}{\partial \theta}(\theta|X) = \sum_{i=1}^n \frac{\partial \log f(X_i|\theta)}{\partial \theta}.$$

By the first order condition for the ML estimator, $S(\hat{\theta}_{ML}) = 0$. By the first information equality,

$$\mathbb{E}[S(\theta_0)] = \sum_{i=1}^n \mathbb{E} \left[\frac{\partial \log f(X_i|\theta_0)}{\partial \theta} \right] = 0.$$

By the definition of Fisher information,

$$\mathbb{E} \left[\left(\frac{\partial \log f(X_i|\theta)}{\partial \theta} \right)^2 \right] = I_1(\theta_0).$$

So, by the central limit theorem, under the null hypothesis (if θ is 1-dimensional)

$$\frac{1}{\sqrt{n}} \frac{S(\theta_0)}{\sqrt{I_1(\theta_0)}} \xrightarrow{d} N(0, 1).$$

By the continuous mapping theorem,

$$LM = S(\theta_0)^2 / (nI_1(\theta_0)) \xrightarrow{d} \chi_1^2.$$

The LM is called the Lagrange multiplier (LM) statistic. Let us show where the name comes from. Consider the constrained optimization problem $\log \mathcal{L}(\theta|x) \rightarrow \max$ s.t. $\theta = \theta_0$. The Lagrangian is

$$H = \log \mathcal{L}(\theta|x) - \lambda(\theta - \theta_0).$$

The FOC is

$$S(\theta_0) = \lambda.$$

So, indeed, the score is connected to the Lagrange multiplier.

Comparison to LR and Wald Let us show that $LM - LR \rightarrow_p 0$. By Taylor's expansion,

$$S(\theta_0) = S(\theta_0) - S(\hat{\theta}_{ML}) = \frac{\partial^2 \ell}{\partial \theta^2}(\theta^*|X)(\theta_0 - \hat{\theta}_{ML}),$$

where θ^* is between θ_0 and $\hat{\theta}_{ML}$. As before, $-(1/n)\partial^2 \ell(\theta^*|X)/\partial \theta^2 \rightarrow_p I_1(\theta_0)$. By the Slutsky theorem,

$$LM = n^2 \left(\frac{1}{n} \frac{\partial^2 \ell}{\partial \theta^2}(\theta^*|X) \right)^2 (\theta_0 - \hat{\theta}_{ML})^2 / (nI_1(\theta_0)) = nI_1(\theta_0)(\theta_0 - \hat{\theta}_{ML})^2(1 + o_p(1))$$

Thus, we have shown that LR, Wald, and LM statistics are all asymptotically equivalent under the null hypothesis. However, they differ in finite samples. For example, in the case of normal likelihood, we have $LM \leq LR \leq W$.

3.2 Multi-dimensional case

Assume that the unknown parameter θ is k -dimensional, while the null hypothesis is imposing p -dimensional restriction $\Theta_0 = \{\theta \in \Theta : g_1(\theta) = 0, \dots, g_p(\theta) = 0\}$. The score function is the $k \times 1$ vector function $S(\theta) = \frac{\partial \ell}{\partial \theta}(\theta|X)$. Denote $\hat{\theta}_0$ to be the restricted estimator: $\hat{\theta}_0 = \arg \max_{\theta \in \Theta_0} \ell(\theta|X)$. Then

$$LM = \frac{1}{n} S(\hat{\theta}_0)' I_1(\hat{\theta}_0)^{-1} S(\hat{\theta}_0) \xrightarrow{d} \chi_p^2$$

if the null holds.

An advantage of the LM statistic is that it only includes calculations based on the restricted estimator $\hat{\theta}_0$. On the other hand, in order to find the LM statistic, we have to estimate the Fisher information.

Example (cont.) Let us calculate the LM statistic in our example with a random sample from the Poisson(λ) distribution. We have

$$S(\lambda_0) = \sum_{i=1}^n X_i/\lambda_0 - n = 500/6 - 100 = -100/6$$

and $I_1(\lambda_0) = 1/\lambda_0 = 1/6$. So,

$$LM = \frac{S(\lambda_0)^2}{n I_1(\lambda_0)} = \frac{1}{100} \cdot \left(\frac{100}{6}\right)^2 \cdot 6 = \frac{100}{6} \approx 17$$

4 Generalizations and Summary

Let $X = (X_1, \dots, X_n)$ be a random sample from a distribution $f(x|\theta)$ with $\theta \in \Theta$. Suppose we want to test the null hypothesis, H_0 , that $\theta \in \Theta_0$ against the alternative hypothesis, H_a , that $\theta \notin \Theta_0$. Let $\hat{\theta}_0$ be a restricted estimator, i.e. $\hat{\theta}_0$ solves $\max_{\theta \in \Theta_0} \mathcal{L}(\theta|X)$, and $\hat{\theta}_{ML}$ an unrestricted (ML) estimator, i.e. $\hat{\theta}_{ML}$ solves $\max_{\theta \in \Theta} \mathcal{L}(\theta|X)$. Assume for simplicity that the null can be formulated as $g(\theta) = 0$, where g is a p -dimensional function. Then, under the null hypothesis,

$$LR = 2(\ell(\hat{\theta}_{ML}|X) - \ell(\hat{\theta}_0|X)) \xrightarrow{d} \chi_p^2$$

$$W = n(g(\hat{\theta}_{ML}) - 0)' \hat{\Sigma}^{-1} (g(\hat{\theta}_{ML}) - 0) \xrightarrow{d} \chi_p^2$$

$$LM = S(\hat{\theta}_0)' I_n^{-1}(\hat{\theta}_0) S(\hat{\theta}_0) \xrightarrow{d} \chi_p^2$$

where $\hat{\Sigma} = \left(\frac{\partial g}{\partial \theta}(\hat{\theta}_{ML})\right)' I_1^{-1}(\hat{\theta}_{ML}) \left(\frac{\partial g}{\partial \theta}(\hat{\theta}_{ML})\right)$ is a natural delta-method inspired estimate of the asymptotic variance. Under proper regularity conditions all of these tests are asymptotically equivalent to each other.

Notice that LR and LM are invariant to the formulation of the null hypothesis, while Wald is not.